DATA STRUCTURES

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HOME WORK:1

1.Write a C program to find the **sum of two matrices**.

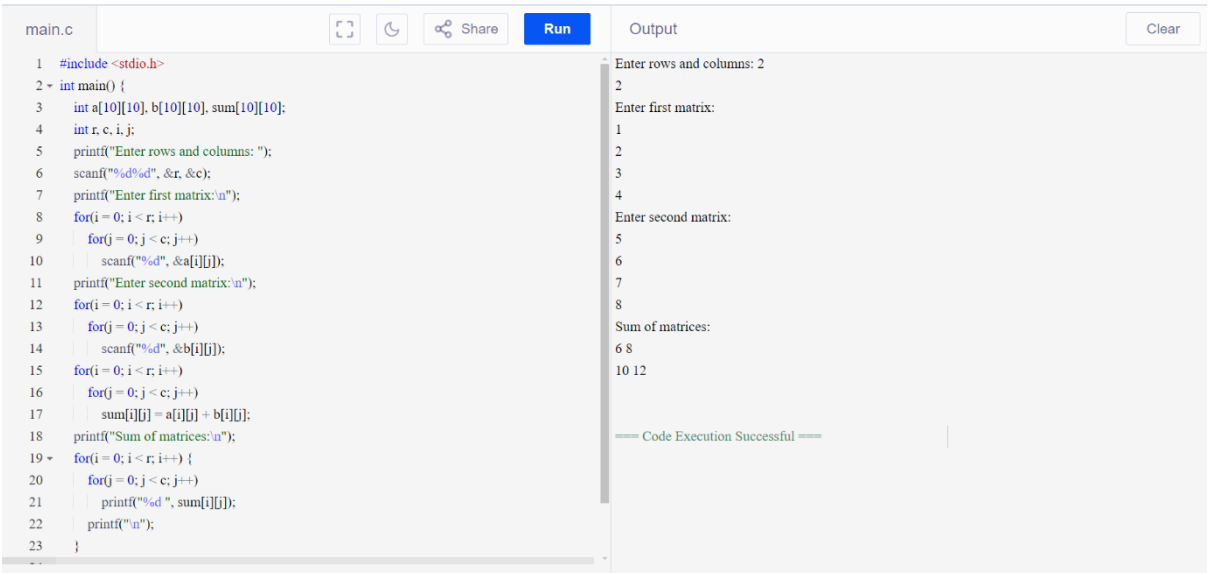
CODE:

```
#include <stdio.h>
int main() {
    int a[10][10], b[10][10], sum[10][10];
    int r, c, i, j;
    printf("Enter rows and columns: ");
    scanf("%d%d", &r, &c);
    printf("Enter first matrix:\n");
    for(i = 0; i < r; i++)
        for(j = 0; j < c; j++)
            scanf("%d", &a[i][j]);
    printf("Enter second matrix:\n");
    for(i = 0; i < r; i++)
        for(j = 0; j < c; j++)
            scanf("%d", &b[i][j]);
    for(i = 0; i < r; i++)
        for(j = 0; j < c; j++)
            sum[i][j] = a[i][j] + b[i][j];
    printf("Sum of matrices:\n");
    for(i = 0; i < r; i++) {
        for(j = 0; j < c; j++)
```

```
            printf("%d ", sum[i][j]);
        printf("\n");
    }

    return 0;
}
```

OUTPUT:



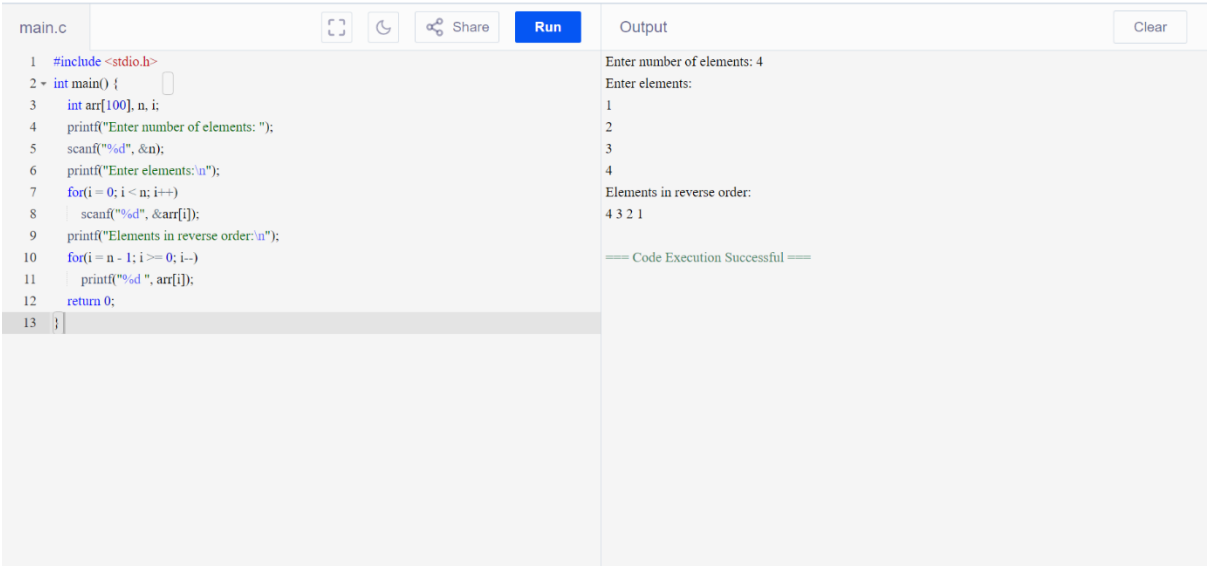
2.Write a C program to read **N numbers into an array** and display them in **reverse order**.

CODE:

```
#include <stdio.h>

int main() {
    int arr[100], n, i;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);
```

```
    printf("Elements in reverse order:\n");
    for(i = n - 1; i >= 0; i--)
        printf("%d ", arr[i]);
    return 0;
}
```



3.Write a C program to count the **total number of duplicate elements** in an array.

CODE:

```
#include <stdio.h>
int main() {
    int arr[100], n;
    int i, j, count = 0;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    for(i = 0; i < n; i++) {
        for(j = i + 1; j < n; j++) {
```

```
        if(arr[i] == arr[j]) {
            count++;
            break;
        }
      }
   }
   printf("Total duplicate elements = %d", count);
   return 0;
}
```

OUTPUT:



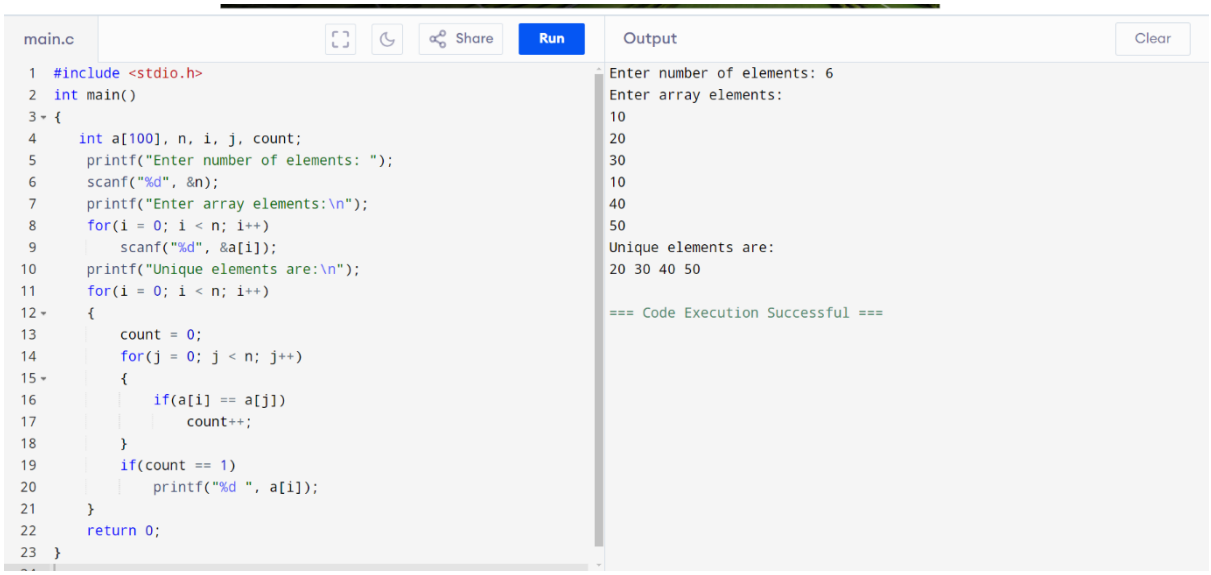
4. Write a program to print all the elements unique elements in an array.

CODE:

```
#include <stdio.h>
int main()
{
  int a[100], n, i, j, count;
   printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
    printf("Unique elements are:\n");
    for(i = 0; i < n; i++)
    {
        count = 0;
        for(j = 0; j < n; j++)
        {
            if(a[i] == a[j])
                count++;
        }
        if(count == 1)
            printf("%d ", a[i]);
    }
    return 0;
}
```

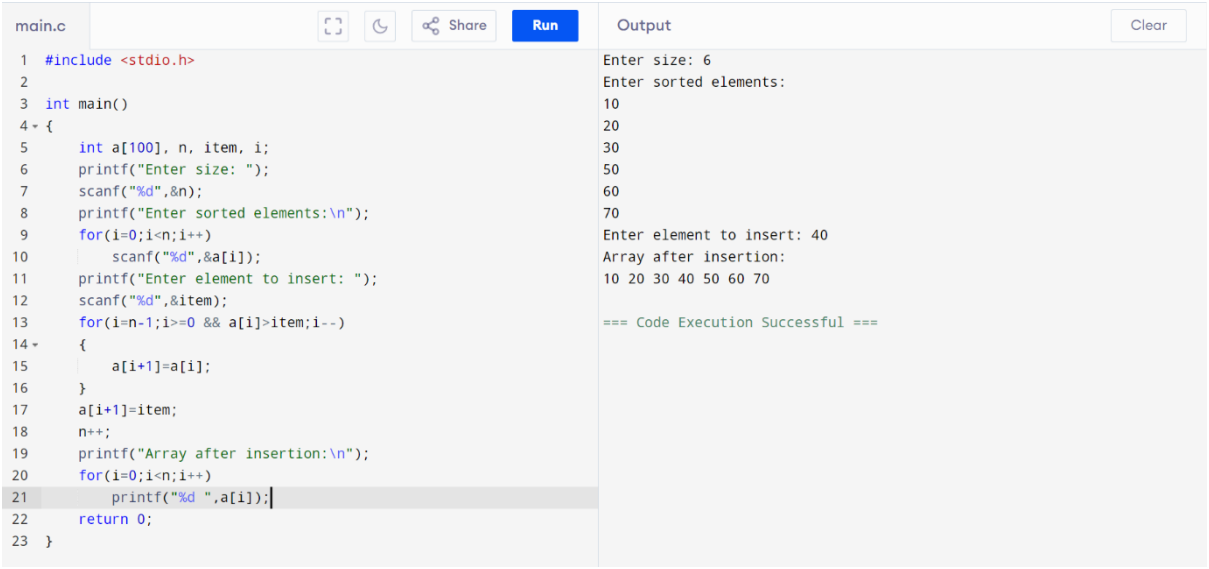
OUTPUT:



5.Write a c program to insert an element in a sorted array.

```
#include <stdio.h>
int main()
{
   int a[100], n, item, i;
   printf("Enter size: ");
   scanf("%d",&n);
   printf("Enter sorted elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&a[i]);
   printf("Enter element to insert: ");
   scanf("%d",&item);
   for(i=n-1;i>=0 && a[i]>item;i--)
   {
      a[i+1]=a[i];
   }
   a[i+1]=item;
   n++;
   printf("Array after insertion:\n");
   for(i=0;i<n;i++)
      printf("%d ",a[i]);
   return 0;
}
```

OUTPUT:



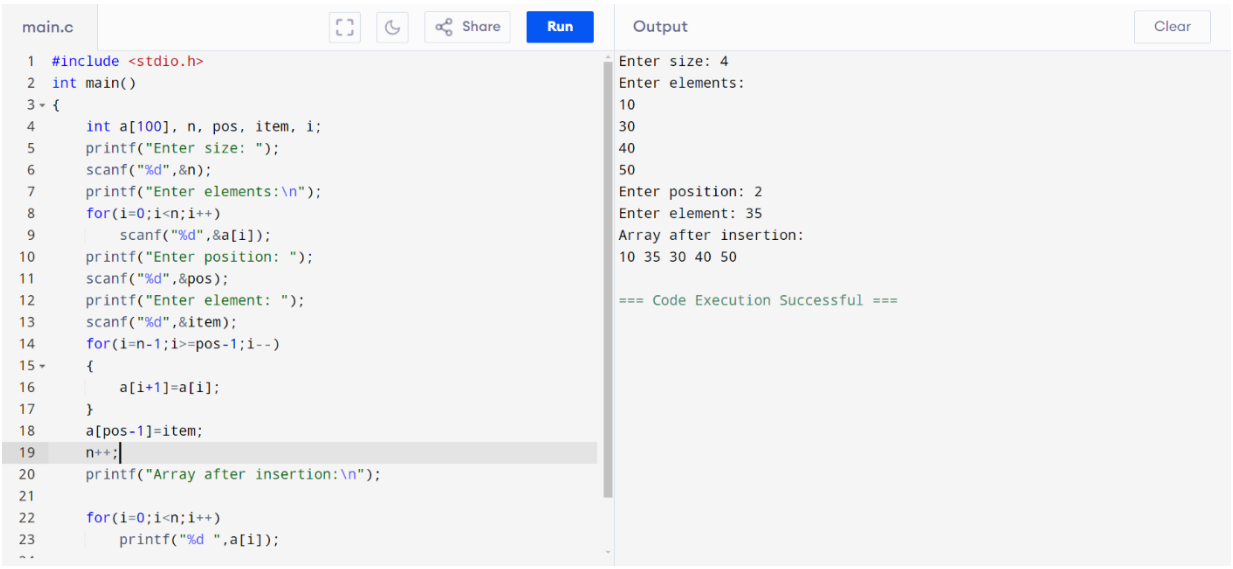
6.Write a c program to insert an element in an unsorted array.

CODE:

```
#include <stdio.h>
int main()
{
   int a[100], n, pos, item, i;
   printf("Enter size: ");
   scanf("%d",&n);
   printf("Enter elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&a[i]);
   printf("Enter position: ");
   scanf("%d",&pos);
   printf("Enter element: ");
   scanf("%d",&item);
   for(i=n-1;i>=pos-1;i--)
   {
      a[i+1]=a[i];
```

```
    }
    a[pos-1]=item;
    n++;
    printf("Array after insertion:\n");
    for(i=0;i<n;i++)
        printf("%d ",a[i]);
    return 0;
}
```

OUTPUT:



7.Write a c program to perform matrix multiplication.

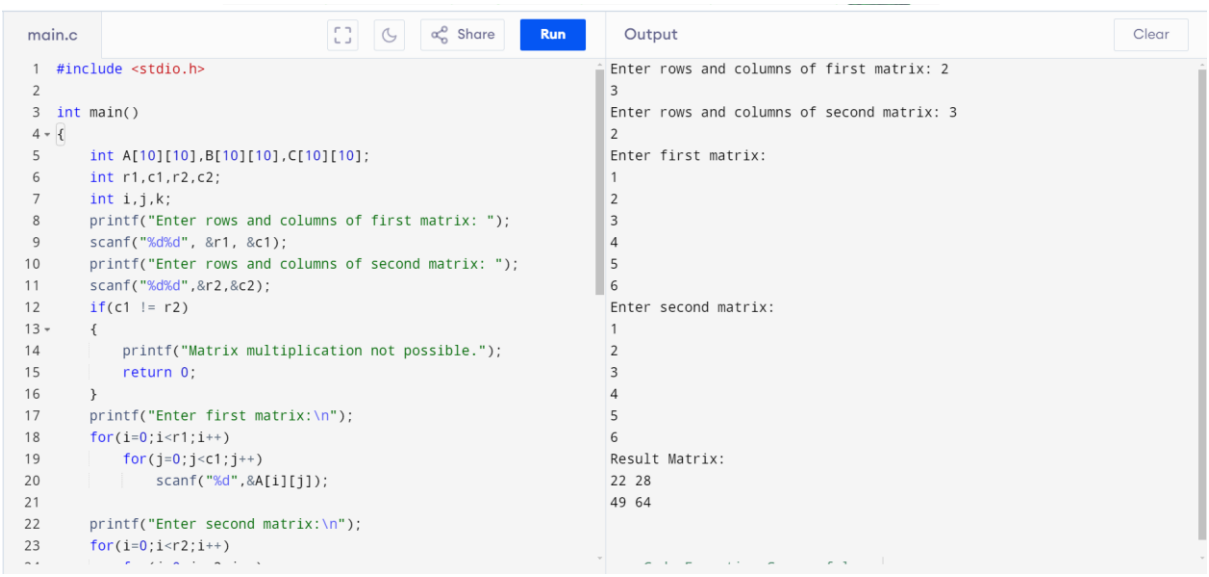
CODE:

```
#include <stdio.h>
int main()
{
    int A[10][10],B[10][10],C[10][10];
    int r1,c1,r2,c2;
    int i,j,k;
```

```
printf("Enter rows and columns of first matrix: ");
scanf("%d%d", &r1, &c1);
printf("Enter rows and columns of second matrix: ");
scanf("%d%d",&r2,&c2);
if(c1 != r2)
{
   printf("Matrix multiplication not possible.");
   return 0;
}
printf("Enter first matrix:\n");
for(i=0;i<r1;i++)
   for(j=0;j<c1;j++)
      scanf("%d",&A[i][j]);
printf("Enter second matrix:\n");
for(i=0;i<r2;i++)
   for(j=0;j<c2;j++)
      scanf("%d",&B[i][j]);
for(i=0;i<r1;i++)
{
   for(j=0;j<c2;j++)
   {
      C[i][j]=0;
      for(k=0;k<c1;k++)
      {
         C[i][j]+=A[i][k]*B[k][j];
      }
   }
}
```

```
    printf("Result Matrix:\n");
    for(i=0;i<r1;i++)
    {
        for(j=0;j<c2;j++)
            printf("%d ",C[i][j]);
        printf("\n");
    }
    return 0;
}
```

OUTPUT:



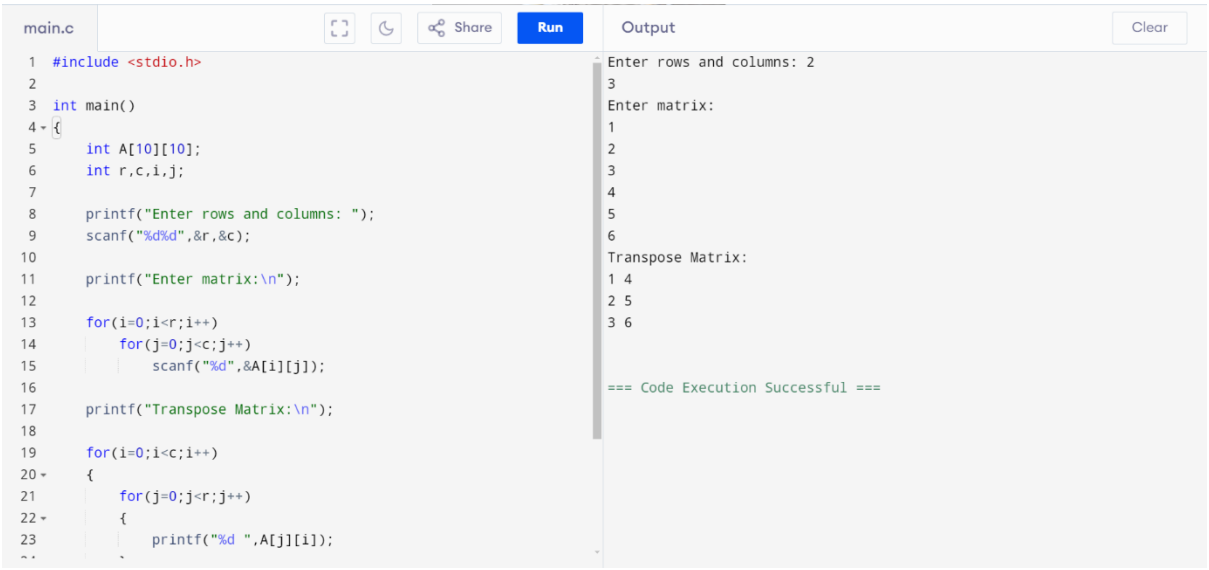
8.Write a c program to perform traverse of a matrix.

CODE:

```
#include <stdio.h>
int main()
{
    int A[10][10];
    int r,c,i,j;
    printf("Enter rows and columns: ");
    scanf("%d%d",&r,&c);
```

```
    printf("Enter matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&A[i][j]);
    printf("Transpose Matrix:\n");
    for(i=0;i<c;i++)
    {
        for(j=0;j<r;j++)
        {
            printf("%d ",A[j][i]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:



9.Write a c program to print the sum of elements in the right diagonal in the matrix.

CODE:

#include <stdio.h>

```
int main()
{
   int A[10][10];
   int n,i,j,sum=0;
   printf("Enter order of square matrix: ");
   scanf("%d",&n);
   printf("Enter matrix:\n");
   for(i=0;i<n;i++)
      for(j=0;j<n;j++)
         scanf("%d",&A[i][j]);
   for(i=0;i<n;i++)
   {
      for(j=0;j<n;j++)
      {
         if(i+j==n-1)
            sum+=A[i][j];
      }
   }
   printf("Right diagonal sum = %d",sum);
   return 0;
}
```

OUTPUT:



10.Write a c program to check for identity matrix.

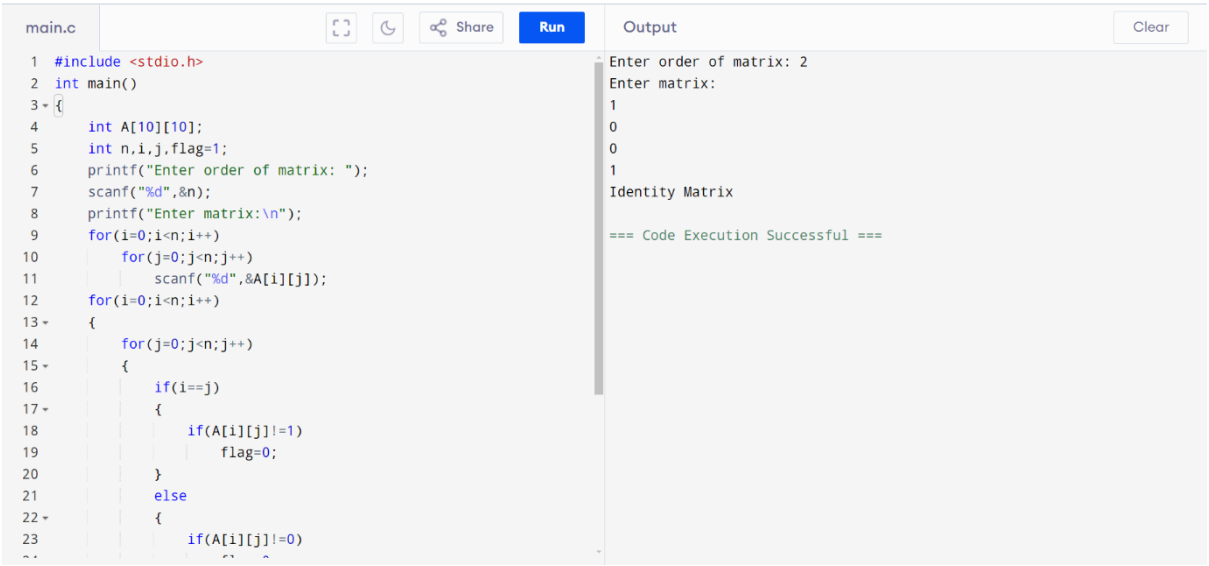
CODE:

```
#include <stdio.h>
int main()
{
   int A[10][10];
   int n,i,j,flag=1;
   printf("Enter order of matrix: ");
   scanf("%d",&n);
   printf("Enter matrix:\n");
   for(i=0;i<n;i++)
      for(j=0;j<n;j++)
         scanf("%d",&A[i][j]);
   for(i=0;i<n;i++)
   {
      for(j=0;j<n;j++)
      {
         if(i==j)
         {
```

```
            if(A[i][j]!=1)
                flag=0;
        }
        else
        {
            if(A[i][j]!=0)
                flag=0;
        }
    }
}
if(flag==1)
    printf("Identity Matrix");
else
    printf("Not an Identity Matrix");

return 0;
}
```

OUTPUT:



11.Write a c program to find the number of vowels, consonants, whitespaces, and digits.

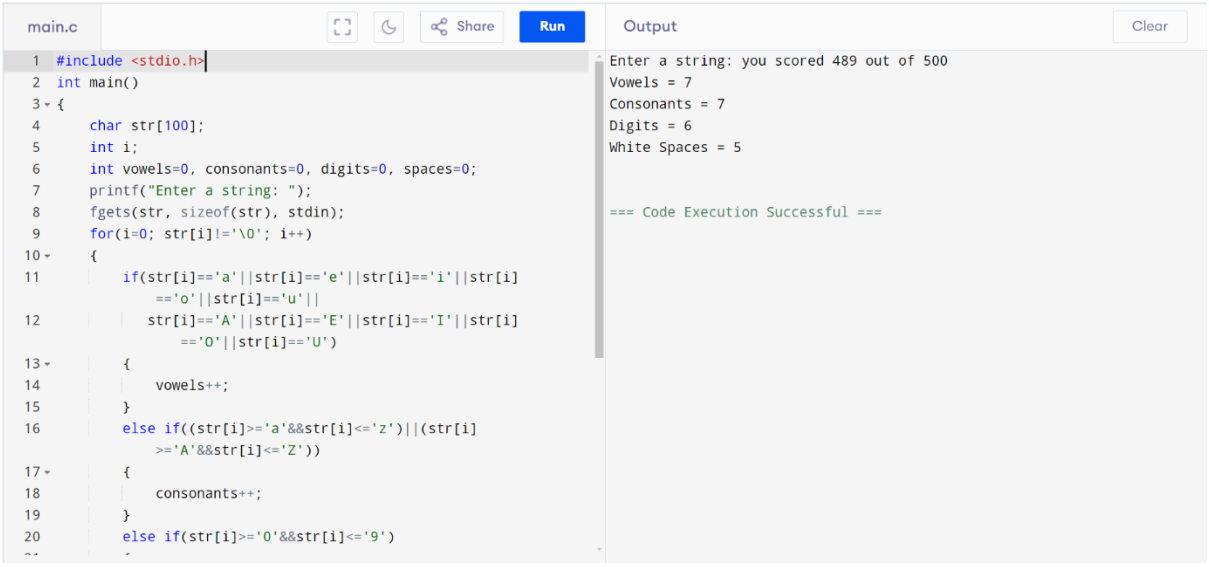
CODE:

```
#include <stdio.h>
int main()
{
   char str[100];
   int i;
   int vowels=0, consonants=0, digits=0, spaces=0;
   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);
   for(i=0; str[i]!='\0'; i++)
   {
      if(str[i]=='a'||str[i]=='e'||str[i]=='i'||str[i]=='o'||str[i]=='u'||
        str[i]=='A'||str[i]=='E'||str[i]=='I'||str[i]=='O'||str[i]=='U')
      {
         vowels++;
      }
      else if((str[i]>='a'&&str[i]<='z')||(str[i]>='A'&&str[i]<='Z'))
      {
         consonants++;
      }
      else if(str[i]>='0'&&str[i]<='9')
      {
         digits++;
      }
      else if(str[i]==' ')
      {
         spaces++;
```

```
    }
  }
  printf("Vowels = %d\n", vowels);
  printf("Consonants = %d\n", consonants);
  printf("Digits = %d\n", digits);
  printf("White Spaces = %d\n", spaces);

  return 0;
}
```

OUTPUT:



12.Write a c program to print the frequency of a character in a string.

CODE:

```
#include <stdio.h>
int main()
{
  char str[100];
  int i, j, count;
  printf("Enter a string: ");
  fgets(str, sizeof(str), stdin);
```

```
    for(i=0; str[i]!='\0'; i++)
    {
        if(str[i]==' ')
            continue;
        count=1;
        if(str[i]=='*')
            continue;
        for(j=i+1; str[j]!='\0'; j++)
        {
            if(str[i]==str[j])
            {
                count++;
                str[j]='*';
            }
        }
        printf("%c = %d\n",str[i],count);
    }
    return 0;
}
```

OUTPUT:

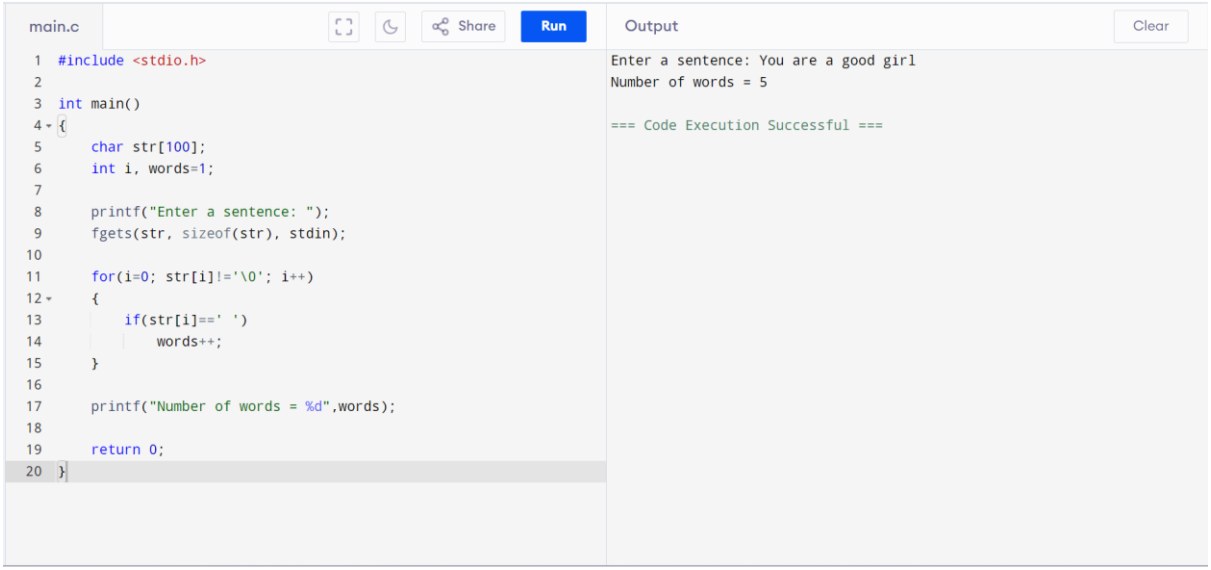


13.Write a c program to count the number of words in a string.

CODE:

```
#include <stdio.h>
int main()
{
   char str[100];
   int i, words=1;
   printf("Enter a sentence: ");
   fgets(str, sizeof(str), stdin);
   for(i=0; str[i]!='\0'; i++)
   {
      if(str[i]==' ')
         words++;
   }
   printf("Number of words = %d",words);
   return 0;
}
```

OUTPUT:



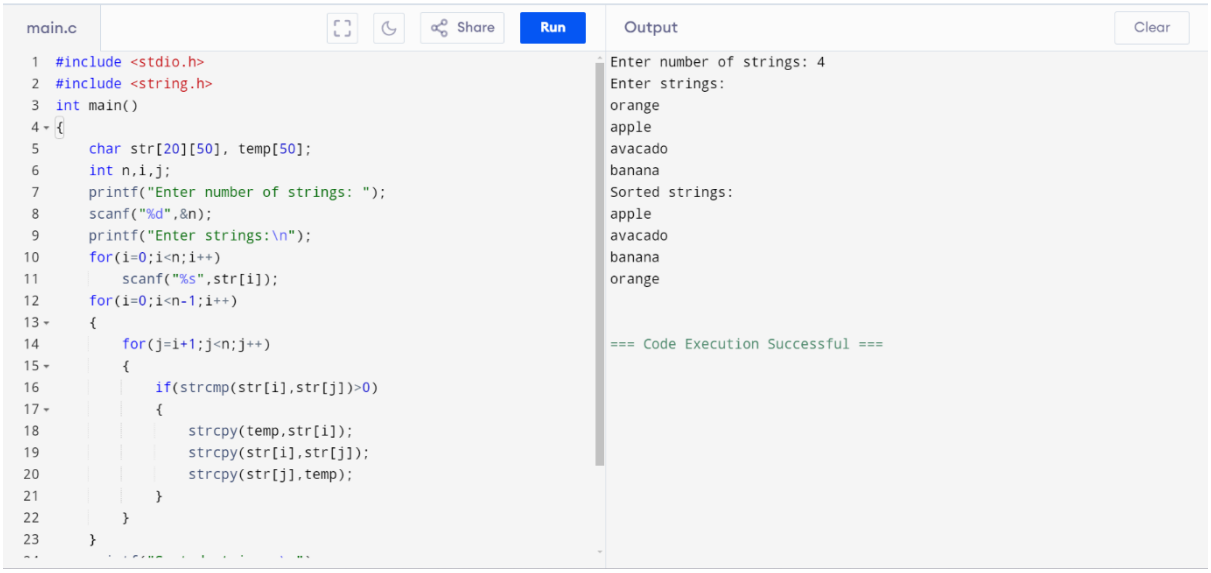
14.Write a c program to perform sorting of string array.

CODE:

```
#include <stdio.h>
#include <string.h>
int main()
{
   char str[20][50], temp[50];
   int n,i,j;
   printf("Enter number of strings: ");
   scanf("%d",&n);
   printf("Enter strings:\n");
   for(i=0;i<n;i++)
      scanf("%s",str[i]);
   for(i=0;i<n-1;i++)
   {
      for(j=i+1;j<n;j++)
      {
         if(strcmp(str[i],str[j])>0)
         {
```

```
            strcpy(temp,str[i]);
            strcpy(str[i],str[j]);
            strcpy(str[j],temp);
        }
    }
}
printf("Sorted strings:\n");
for(i=0;i<n;i++)
    printf("%s\n",str[i]);
return 0;
}
```

OUTPUT:



15.Write a c program to toggle the given string.

CODE:

```
#include <stdio.h>
int main()
{
    char str[100];
    int i;
```

```
    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);
    for(i=0; str[i]!='\0'; i++)
    {
        if(str[i]>='A' && str[i]<='Z')
            str[i]=str[i]+32;
        else if(str[i]>='a' && str[i]<='z')
            str[i]=str[i]-32;
    }
    printf("After Toggle Case:\n%s",str);
    return 0;
}
```

OUTPUT:

